

Characterisation of a multi-user ultrasound-guided focused ultrasound system.

Supervisors:

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Aim: To experimentally characterise a state-of-the art ultrasound-guided ultrasound therapy system and establish treatment guidelines for different therapeutic applications, including neuromodulation, tissue ablation, and opening the blood-brain barrier.

Project description: As part of a recent EPSRC / faculty call for multi-user equipment, we purchased a state-of-the art ultrasound-guided ultrasound therapy system (installed in March 2022). The system can deliver ultrasound energy, for example, for tissue ablation, drug delivery, and neuromodulation. This is a multi-user system, with a current consortium of 26 interested users across 9 UCL departments (including members of WEISS). The equipment is hosted and managed by the Department of Medical Physics and Biomedical Engineering, and is accessible to all staff.

The system delivers ultrasound energy using one of two transducers: a low frequency transducer designed for human or large animal applications, and a high frequency transducer designed for small animal applications. In both cases, targeting is performed under ultrasound guidance using an integrated graphical user interface (GUI). The GUI allows setting the ultrasound therapy parameters, including the duty cycle, pulse length, and driving voltage. However, the best parameters to use for a given therapy are currently not known.

The goal of this project is to perform a systematic experimental calibration of the system for both transducers and create a user guide. The user guide will be deployed in the form of a wiki accessible to end users.

Research: The calibration of the system is essential before end-users can use the system safely and effectively. The data gathered in this project will therefore underpin the many anticipated applications likely to generate significant impact. We have a good relationship with the company who manufactures the system (Verasonics), and they will offer their support in the way of training, as well as potential modifications of the GUI (e.g., to display derived parameters or calibration information).

Once calibrated, specific impact from the deployment of the system could include: (i) providing new fundamental discoveries on the bioeffects of ultrasound at the cellular level, (ii) accelerating the clinical translation of interventions for cancer, pain, heart disease, and neurological disorders, (iii) developing new ultrasound-based technologies for organ preservation, functional brain imaging, tissue fractionation, and shaping acoustic waves, (iv) disseminating new knowledge through publications, conference presentations, and public engagement activities, (v) influencing public policy, e.g., related to minimally invasive procedures, (vi) generating new income through patents, commercialisation activities, and by providing pilot data for grant applications, and (vii) facilitating new collaborations across the consortium by providing a platform for communication between UCL researchers interested in ultrasound therapy.

Work plan:

Goal 1: Training (1 week). The student will be trained in the use of the ultrasound and measurement systems.

Goal 2: Establish the relationship between the driving voltage and acoustic pressure (2 weeks). Hydrophone measurements will be made in a precision scanning tank with a calibrated hydrophone to establish how the focal pressure varies with the driving conditions.

Goal 3: Establish the relationship between the driving voltage and acoustic power (2 weeks). Power measurements will be made using a calibrated radiation force balance.

Goal 4: Establish the relationship between driving parameters and tissue ablation (2 weeks). Systematic ablation experiments will be conducted using tissue mimics and ex vivo animal tissue to establish how the size of the ablated volume changes with the driving parameters.

Goal 5: Establish treatment guidelines (1 week). Using the results from the experimental measurements, a set of treatment guidelines will be established outlining how the driving parameters should be set for different anticipated applications of the system (including neuromodulation, tissue ablation, and opening the blood-brain barrier). This information will be deployed in an open wiki accessible to users of the system.

Benefits to the student: Through the project work and training provided, the student will gain skills and experience in the characterisation of medical ultrasound systems, measurement of ultrasound fields, test and measurement procedures and instrumentation. These skills will be a useful foundation for further work in ultrasound research or for experimental physics or engineering work in general. The student will also gain experience of signal processing using matlab and relevant toolboxes. They will be working within a supportive team with regular 1-1 and project meetings where support, training and direction will be provided. There will be opportunities to find out about the different areas of research and collaborations for which the ultrasound system will be used, and to gain experiencing of liaising and working with industry (the equipment manufacturers).